

Generative AI Engineering & Agentic AI Development Program

Total Duration: ~80 Hours

Session Objective (SO)

Enable participants to design, build, and deploy production-ready Generative AI and Agentic AI systems by understanding foundation models, developing LLM applications, implementing retrieval pipelines, designing autonomous AI agents, and deploying scalable AI solutions using modern frameworks and cloud infrastructure.

Learning Objectives (LO)

By the end of this program, participants will be able to:

- Understand the architecture and capabilities of modern Generative AI and foundation models
 - Build applications using LLM APIs, prompt engineering, and structured outputs
 - Develop full-stack AI applications integrating backend services and user interfaces
 - Implement embeddings, vector databases, and Retrieval Augmented Generation (RAG) systems
 - Build multimodal AI systems using text, image, and audio models
 - Understand how Agentic AI differs from traditional Generative AI applications
 - Design and implement AI agents using LangGraph and CrewAI frameworks
 - Apply agentic design patterns for planning, reasoning, and orchestration
 - Implement agent-to-agent communication protocols and MCP-based integrations
 - Containerize and deploy AI applications using Docker and cloud infrastructure
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Module 1: Foundations of Generative AI and LLMs

Duration: 6 Hours

Module Outline

- Evolution of AI, Machine Learning, Deep Learning, and Generative AI
- Foundation models and transformer architecture
- Tokenization, embeddings, and context windows
- Overview of the LLM ecosystem (OpenAI, Claude, Gemini, Llama, Mistral)
- Multimodal capabilities in modern AI systems

Hands-on Activity

Explore multiple LLM APIs and generate outputs from text, image, and audio inputs.

Tools Used

Python
Jupyter Notebook
HuggingFace
OpenAI API

Project

Build a **Multimodal AI Assistant**.

Module 2: LLM APIs, Prompt Engineering, and Function Calling

Duration: 6 Hours

Module Outline

- Introduction to LLM APIs and SDKs
- Calling models using Python and API clients
- Prompt engineering techniques for application development
- Function calling and tool integrations
- Structured outputs using JSON schemas

Hands-on Activity

Build a chatbot that integrates external APIs.

Tools Used

Python
OpenAI API
Gemini API
FastAPI

Project

Build an **AI Personal Assistant with external API integration**.

Module 3: Backend and Frontend Development for AI Applications

Duration: 8 Hours

Module Outline

- Backend development using Python and FastAPI
- Designing AI service endpoints and APIs
- Frontend development using Streamlit
- Building interfaces with HTML and CSS
- Integrating ReactJS and Node.js with AI APIs

Hands-on Activity

Develop a web interface for an AI assistant.

Tools Used

Python
FastAPI

Streamlit
ReactJS
Node.js

Project

Build a **Full-Stack AI Chatbot Application**.

Module 4: LangChain and LLM Orchestration

Duration: 8 Hours

Module Outline

- LangChain architecture and ecosystem
- Prompt templates and chain design
- Memory management for conversational AI
- Agents and tool integrations
- Designing reasoning pipelines

Hands-on Activity

Build a multi-step reasoning assistant using LangChain.

Tools Used

LangChain
Python

Project

Build an **AI Research Assistant**.

Module 5: Embeddings and Vector Databases

Duration: 6 Hours

Module Outline

- Embeddings and semantic similarity concepts
- Vector database fundamentals
- FAISS, Pinecone, Chroma, and Weaviate overview
- Document chunking and indexing strategies
- Retrieval pipelines and metadata filtering

Hands-on Activity

Build a semantic document search system.

Tools Used

FAISS
ChromaDB
HuggingFace Embeddings

Project

Build a **Semantic Knowledge Search Engine**.

Module 6: Retrieval Augmented Generation (RAG)

Duration: 8 Hours

Module Outline

- RAG architecture and knowledge grounding
- Document ingestion and preprocessing pipelines
- Chunking strategies and retrieval techniques

- Combining retrieval with LLM reasoning
- Performance optimization and evaluation

Hands-on Activity

Build a RAG chatbot over PDFs.

Tools Used

LangChain
FAISS
Python

Project

Build a **Enterprise Knowledge Assistant**.

Module 7: Multimodal AI Systems

Duration: 6 Hours

Module Outline

- Vision-language models and multimodal architectures
- Image understanding and captioning pipelines
- Audio processing and speech transcription
- Multimodal retrieval systems
- Building multimodal AI assistants

Hands-on Activity

Build an image and text analysis assistant.

Tools Used

CLIP
Whisper
OpenAI Vision APIs

Project

Build a **Multimodal Document Analysis System**.

Module 8: Generative AI vs Agentic AI Systems

Duration: 4 Hours

Module Outline

- What is Generative AI vs Agentic AI
- Characteristics of autonomous AI agents
- Tool usage and reasoning capabilities
- Planning, execution, and memory in agents
- Real-world agentic AI applications

Hands-on Activity

Convert a simple chatbot into a tool-using AI agent.

Tools Used

Python
LangChain

Project

Build a **Tool-Using AI Assistant**.

Module 9: Agentic AI Design Patterns

Duration: 6 Hours

Module Outline

- Planning and reasoning patterns for AI agents
- ReAct and tool-use design patterns
- Reflection and self-correction patterns
- Multi-step task decomposition patterns
- Orchestration and control-flow patterns

Hands-on Activity

Implement a reasoning agent using ReAct pattern.

Tools Used

LangChain
Python

Project

Build an **AI Planning and Reasoning Agent**.

Module 10: LangGraph for Agent Workflows

Duration: 6 Hours

Module Outline

- LangGraph architecture and graph-based execution
- State management in agent workflows
- Conditional execution and control flow

- Multi-step reasoning pipelines
- Building autonomous AI workflows

Hands-on Activity

Create a planning-based agent workflow using LangGraph.

Tools Used

LangGraph
Python

Project

Build a **Task Automation Agent using LangGraph**.

Module 11: CrewAI for Multi-Agent Systems

Duration: 6 Hours

Module Outline

- CrewAI architecture and design philosophy
- Role-based AI agents and responsibilities
- Task delegation and collaboration between agents
- Multi-agent coordination strategies
- Building AI teams for complex workflows

Hands-on Activity

Build a collaborative multi-agent research system.

Tools Used

CrewAI
Python

Project

Build a **Multi-Agent AI Research Team**.

Module 12: Agent-to-Agent Communication (A2A)

Duration: 4 Hours

Module Outline

- Fundamentals of agent-to-agent communication
- Communication protocols for distributed agents
- Message passing and coordination strategies
- Collaborative reasoning across agents
- Designing distributed AI workflows

Hands-on Activity

Build two communicating AI agents.

Tools Used

LangGraph
CrewAI

Project

Build a **Collaborative AI Agent System**.

Module 13: Model Context Protocol (MCP)

Duration: 4 Hours

Module Outline

- Introduction to Model Context Protocol
- Tool interoperability using MCP
- Integrating external systems through MCP servers
- Building MCP-enabled AI agents
- Enterprise use cases for MCP

Hands-on Activity

Connect an AI agent to external tools using MCP.

Tools Used

Claude MCP Tools
Python

Project

Build an **AI Agent with MCP Tool Integrations**.

Module 14: Containerization and Deployment

Duration: 6 Hours

Module Outline

- Docker fundamentals and containerization
- Containerizing AI applications
- Dependency management for AI systems
- Local deployment pipelines
- Production deployment strategies

Hands-on Activity

Dockerize an AI application.

Tools Used

Docker
Python

Project

Deploy a **Containerized AI Application**.

Module 15: Cloud Deployment using AWS

Duration: 6 Hours

Module Outline

- AWS fundamentals for AI applications
- Using AWS Bedrock for foundation models
- IAM roles and permissions
- Deploying services using ECS and AppRunner
- Monitoring and scaling AI services

Hands-on Activity

Deploy an AI chatbot on AWS.

Tools Used

AWS Bedrock
Docker
ECS
IAM

Project

Deploy a **Production AI System on AWS**.

CAPSTONE Projects

Every student has to make a capstone project and build their own industry grade project and deploy

Project ideas that can be done in sessions.

1. **Smart Home Assistant Bot** – An AI chatbot that helps manage household tasks, reminders, and home maintenance queries.
2. **Banking Query Assistant** – A chatbot that answers questions about loans, savings, credit scores, and banking policies using document retrieval.
3. **Medical Report Explainer** – An AI tool that interprets uploaded medical reports and explains results in simple language.
4. **X-Ray Insight Assistant** – A vision-enabled AI that analyzes X-ray images and provides educational insights about possible findings.
5. **AI Meeting Notes Generator** – A system that converts meeting audio into structured summaries and action items.
6. **Resume Screening Assistant** – An AI recruiter assistant that analyzes resumes and ranks candidates based on job descriptions.
7. **AI Travel Planner** – A chatbot that suggests travel itineraries based on budget, location, and interests.
8. **Personal Productivity Coach** – An AI assistant that tracks tasks and suggests daily productivity improvements.
9. **AI Document Summarization Tool** – A system that summarizes long reports, PDFs, and articles into concise insights.
10. **Car Maintenance Advisor** – A chatbot that helps users diagnose common car issues and suggests maintenance actions.

11. **AI Learning Tutor** – A personalized study assistant that explains concepts and generates practice questions.

12. **Financial Planning Assistant** – An AI advisor that suggests budgeting and savings strategies.

13. **Multi-Agent Startup Advisor** – A system where different AI agents collaborate to generate startup ideas and business plans.

14. **AI News Intelligence Assistant** – A chatbot that summarizes news articles and identifies trends across industries.